Calibrating the Coupon Strike of an Autocallable Note under Local Volatility

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The autocallable note

The note is written on the underlying basket of the Nikkei 225 (NKY), the S&P 500 (SPX) and the Hang Seng Index (HSI).

Note denomination, N	USD 10,000
Issue price	100% of N
Trade date	11 November 2023
Maturity date	2 years after trade date
Observation dates, t_m	every six months, $m = 1, \ldots, 4$
Maximum rate of interest, r_+	2% per annum
Minimum rate of interest, r_-	0.01% per annum
Knock-out level, B_+	1.10 (110% of initial spot)
Knock-in level, B_-	0.50 (50% of initial spot)

Note structure

$$R_t = \min_k \frac{S_t^k}{S_0^k}, \qquad \text{coupon rate at } t_m = r_+ \mathbf{1}\{R_{t_m} \geq c\} + r_- \mathbf{1}\{R_{t_m} < c\}$$

- S_t^k is the closing level of index k at time t and S_0^k its initial level, so the reference price R_t is the relative level of the laggard underlying.
- c is the coupon strike that we calibrate.
- A knock-out event, $R_{t_m} \geq B_+$, redeems the note for N, and the coupon for that period is still paid.
- After a knock-in event, $\min_t R_t \leq B_-$, the note repays $N \min(1, R_T)$ at maturity T.

Research question

Question

Which coupon strike prices the autocallable note at 98% of its issue price under a local volatility surface?

Finding

1. Among the local variance surfaces, the Dupire local variance interpolated linearly across maturity leaves the smallest part of the traded moneyness window with negative variance.
2. At the contractual rates $r_+ = 2\%$ and $r_- = 0.01\%$, no coupon strike prices the note at 98% of issue.
3. The two-year USD rate was 4.83%, so a coupon of 2% falls well below the rate at which the note is discounted.

The data are option chains for the three indices from Bloomberg as of 11 November 2023 and overnight index swap rates.

At-the-money forward

$$C(K) - P(K) = S_0 e^{-qT} - Ke^{-rT} \quad \xrightarrow{C=P} \quad K_{\mathrm{ATMF}} = S_0 e^{(r-q)T}$$

- The strike is not quoted under the discrete chain, so we locate the tightest interval $[K_1, K_2]$ with $C(K_1) > P(K_1)$ and $C(K_2) < P(K_2)$, which brackets K_{ATMF}.
- Approximating the first derivatives at K_1 by finite differences gives

$$C(K_{\mathrm{ATMF}}) \approx \frac{C(K_1) + P(K_1)}{2}, \qquad K_{\mathrm{ATMF}} \approx \frac{C(K_{\mathrm{ATMF}}) - C(K_1)}{C(K_2) - C(K_1)} (K_2 - K_1) + K_1.$$

Implied volatility and dividend yield

- Implied variance σ^2_{BS} is recovered from the out-of-the-money quote at each strike by bisection on the Black-Scholes price.

- Dividend yield The implied q follows from the forward strike relation $K_{\mathrm{ATMF}} = S_0 e^{(r-q)T}$ under a non-negativity floor,

$$q \approx \max\left(0,\ r - \frac{\ln(K_{\mathrm{ATMF}}/S_0)}{T}\right),$$

and is interpolated across maturity by a cubic spline to give $q(T)$.

Local volatility

$$\sigma^2_{\mathrm{loc}}(K,T) = \frac{\partial C/\partial T}{\frac{1}{2}K^2\,\partial^2 C/\partial K^2}$$

- The local variance follows from the Dupire equation (Dupire, 1994).
- Both derivatives are taken by three-point central differences on the Black-Scholes price in the forward.
- The step is 1 day in maturity and 1% of the strike spacing in strike.

Gatheral's mapping

$$\sigma_{\text{loc}}^2 = \frac{\partial w/\partial T}{1 - \frac{y}{w}\frac{\partial w}{\partial y} + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)\left(\frac{\partial w}{\partial y}\right)^2 + \frac{1}{2}\frac{\partial^2 w}{\partial y^2}}$$

- Alternatively, the fitted implied surface is mapped to local variance through Gatheral (2002).
- In terms of the total variance $w = \sigma_{\text{BS}}^2 T$, the Dupire equation becomes this relation.
- $y = \ln(K/K_{\text{ATMF}})$ is the log forward moneyness.

Fitting the surface

$$\sigma^2(y) = \sigma^2_{\mathrm{ATMF}} + \delta \frac{\tanh(\kappa y)}{\kappa} + \frac{\gamma}{2}\left(\frac{\tanh(\kappa y)}{\kappa}\right)^2$$

- At each maturity, the implied and the local variance are each fitted across log forward moneyness y, with σ^2 for either variance and σ^2_{ATMF} its at-the-money forward value.
- δ is the skew, γ the convexity and κ the celerity.
- Concave curves are dropped since a concave curve inverts the smile and destabilizes the interpolation across maturity (Appendix F.1).
- The fitted curves are interpolated across maturity linearly or by a cubic spline, a piecewise cubic whose first and second derivatives are continuous at every knot.
- $\{\text{implied}, \text{Dupire}, \text{Gatheral}\} \times \{\text{linear}, \text{cubic}\}$ gives six candidate surfaces, plotted in Appendix F.2 to F.7.

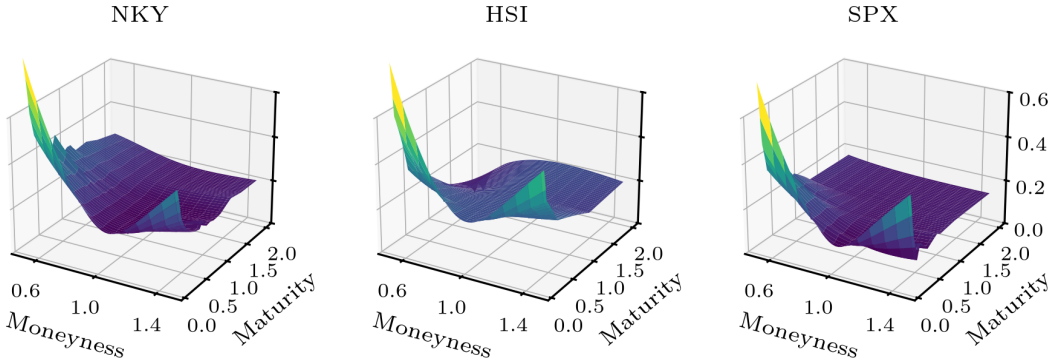
Negative fitted variance

Surface	NKY	HSI	SPX
Implied, cubic	0.0	0.0	10.4
Implied, linear	0.0	0.0	0.0
Dupire, cubic	0.0	12.5	10.6
Dupire, linear	0.0	0.2	0.0
Gatheral, cubic	15.7	14.0	33.6
Gatheral, linear	11.7	10.5	16.1

Percentage of the moneyness window on which the fitted variance is negative.

- Linear interpolation is never worse than cubic, and is strictly better in six of the nine pairs.
- Gatheral's mapping is the weakest of the three, leaving between 10.5% and 33.6% of the window with a negative variance.
- We attribute this to the mapping being applied to an already fitted implied surface rather than to the data.

Selected surface



The selected surface, Dupire local volatility interpolated linearly across maturity. The vertical axis is volatility.

- The simulation employs the Dupire local variance under linear interpolation.
- Linear interpolation joins adjacent maturities by a straight line, so that a positive variance at every knot guarantees a positive variance between them.

Monte Carlo simulation

$$S^k_{t_{j+1}} = S^k_{t_j} \exp\left[\left(\frac{t_{j+1}\, r(t_{j+1}) - t_j\, r(t_j)}{t_{j+1} - t_j} - \tfrac{1}{2}\sigma_j^2\right)(t_{j+1} - t_j) + \sigma_j \sqrt{t_{j+1} - t_j}\, Z^k_j\right],$$

$$\sigma_j^2 = \sigma^2_{\text{loc}}\left(\ln \frac{S^k_{t_j}}{F^k_{t_j}},\ t_j\right), \qquad F^k_T = S^k_0 \exp\{(r(T) - q(T))T\}.$$

- The correlation matrix of SPX, HSI and NKY is computed from the latest 10 years of historical returns.
- A Cholesky factorization then yields three correlated Gaussian processes $\{Z^k\}$ from three independent Wiener processes $\{W^k\}$.
- A negative interpolated σ_j^2 is replaced by the smallest non-negative value seen so far in the simulation (Appendix E.1).

Autocallable pricing

$$V(c) = \frac{1}{M} \sum_{i=1}^{M} (I_i + O_i + G_i)$$

$$I = N \sum_{m=1}^{4} \left(\frac{r_+}{2} \mathbf{1}\{R_{t_m} \geq c\} + \frac{r_-}{2} \mathbf{1}\{R_{t_m} < c\} \right) \Pi_m D_m, \qquad O = N \mathbf{1}\{\tau < \infty\} e^{-r(\tau)\tau},$$

$$G = N \mathbf{1}\{\tau = \infty\} \left(\mathbf{1}\{\min_t R_t \leq B_-\} \min(1, R_T) + \mathbf{1}\{\min_t R_t > B_-\} \right) D_4.$$

- $V(c)$ is the value of the note at coupon strike c, averaged over the M simulated paths.
- I is the variable interest, O the knock-out redemption and G the final redemption.
- $\Pi_m = \prod_{l<m} \mathbf{1}\{R_{t_l} < B_+\}$ is the indicator that no knock-out event has occurred before t_m, and $D_m = e^{-r(t_m)t_m}$ is the discount factor.
- $\tau = \min\{t_m : R_{t_m} \geq B_+\}$ is the first observation date at which the barrier is breached.

Coupon strike optimization

$$V(c) = 0.98N$$

- We calibrate the coupon strike c at which the note prices at 98% of its issue price by bisection, which returns no result if it exhausts 100 iterations.
- At the contractual rates $r_+ = 2\%$ and $r_- = 0.01\%$ the search returns no result.
- We attribute the failure to the level of interest rates. The 2-year USD rate was 4.83%, so a coupon of 2% falls well below the rate at which the note is discounted.
- Since the contractual rates admit no solution, we treat the two rates as free and identify the pairs that do.

Coupon strike at 98% of face

Maximum	Minimum interest, basis points														
interest, %	1.0	8.1	15.1	22.2	29.3	36.4	43.4	50.5	57.6	64.6	71.7	78.8	85.9	92.9	100.0
2.00	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
3.29	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
4.57	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
5.86	0.702	0.703	0.704	0.706	0.707	0.708	0.709	0.711	0.712	0.713	0.715	0.716	0.718	0.719	0.720
7.14	0.798	0.799	0.800	0.802	0.803	0.804	0.805	0.807	0.809	0.810	0.812	0.813	0.815	0.816	0.818
8.43	0.846	0.847	0.849	0.850	0.851	0.853	0.854	0.856	0.857	0.859	0.861	0.862	0.864	0.865	0.867
9.71	0.878	0.879	0.880	0.882	0.883	0.884	0.886	0.887	0.889	0.890	0.892	0.893	0.895	0.896	0.898
11.00	0.901	0.902	0.903	0.905	0.906	0.907	0.909	0.910	0.911	0.913	0.914	0.915	0.917	0.918	0.920
12.29	0.918	0.920	0.921	0.922	0.923	0.925	0.926	0.927	0.929	0.930	0.931	0.933	0.934	0.936	0.937
13.57	0.933	0.934	0.935	0.937	0.938	0.939	0.940	0.941	0.943	0.944	0.946	0.947	0.948	0.950	0.952
14.86	0.945	0.946	0.948	0.949	0.950	0.951	0.953	0.954	0.955	0.956	0.958	0.959	0.961	0.962	0.964
16.14	0.956	0.957	0.958	0.959	0.961	0.962	0.963	0.965	0.966	0.967	0.969	0.970	0.971	0.972	0.974
17.43	0.965	0.966	0.968	0.969	0.970	0.971	0.972	0.973	0.975	0.976	0.977	0.979	0.980	0.981	0.983
18.71	0.973	0.974	0.976	0.977	0.978	0.979	0.980	0.981	0.983	0.984	0.985	0.986	0.988	0.989	0.991
20.00	0.981	0.982	0.983	0.984	0.985	0.986	0.987	0.989	0.990	0.991	0.993	0.994	0.995	0.997	0.998

Coupon strike that prices the note at 98% of face.

A dash marks a pair of coupon rates for which the search returns no result.

At a fixed price, the coupon strike increases with both the maximum and minimum interest.

Conclusion

- Of the four local variance surfaces, the Dupire local variance interpolated linearly across maturity left the smallest part of the traded moneyness window with negative variance.
- Under it, we simulated NKY, SPX and HSI jointly and valued the note, whose coupon, knock-out and knock-in all reference the laggard index.
- At the contractual $r_+ = 2\%$ and $r_- = 0.01\%$, no coupon strike prices the note at 98% of issue, because a coupon well below the discount rate leaves $V(c) < 0.98N$ at any c.
- Treating the two rates as free, the coupon strike increases with both r_+ and r_-.

References

- Dupire, B. (1994). Pricing with a smile. *Risk*, 7, 18–20.
- Gatheral, J. (2002). Lecture 1: Stochastic volatility and local volatility. Case Studies in Financial Modelling Course Notes, Courant Institute of Mathematical Sciences, Fall Term.

Code and replication data are available at *GitHub*.

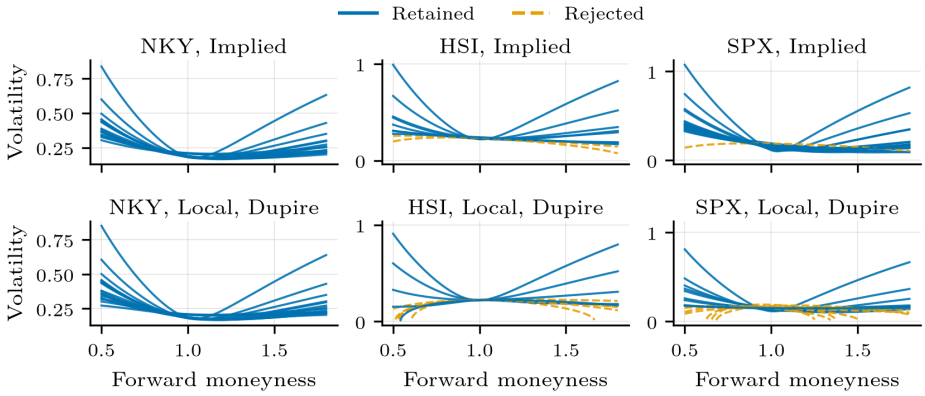
Appendix E.1. Simulation algorithm

Given M paths and J steps, each path proceeds as follows.

1. Evolve the three indices by the recursion of the simulation and form the reference price R_t at every step.
2. Record the knock-in indicator from the running minimum of R_t over all steps, not only the observation dates.
3. At each observation date t_m, accrue the interest if the note is still alive, and record a knock-out at the first date with $R_{t_m} \geq B_+$.
4. Discount each component at its own payment date rather than at maturity, since the knock-out redemption and the coupons are paid when they occur.

Averaging over paths gives $V(c)$, and the coupon strike is the value at which $V(c) = 0.98N$.

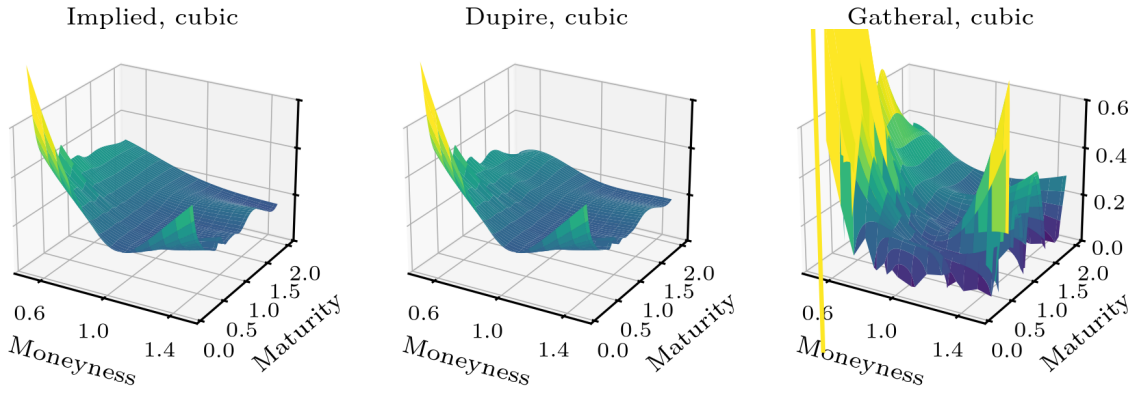
Appendix F.1. Fitted volatility curves



Fitted implied and local volatility curves across forward moneyness.

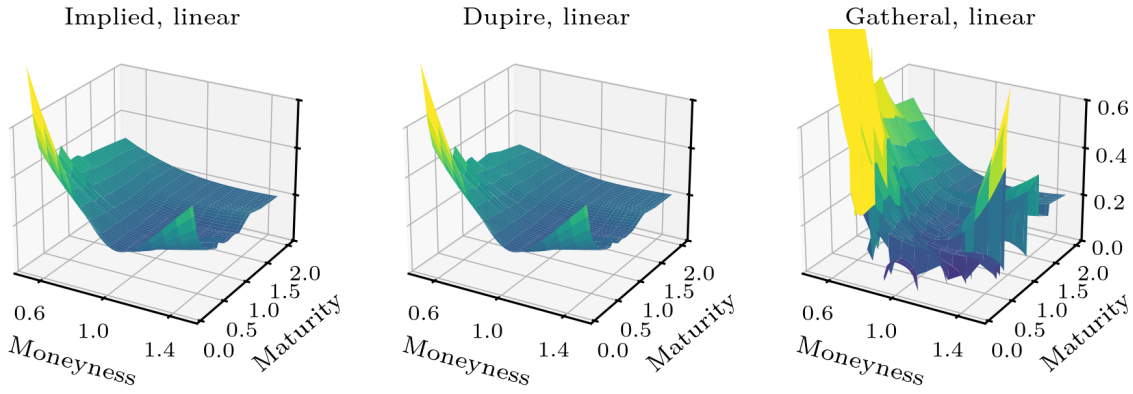
- Curves that are concave are dropped.
- HSI loses two implied and five local curves, SPX one implied and seven local curves, and NKY none.

Appendix F.2. Nikkei 225 fitted surfaces, cubic interpolation



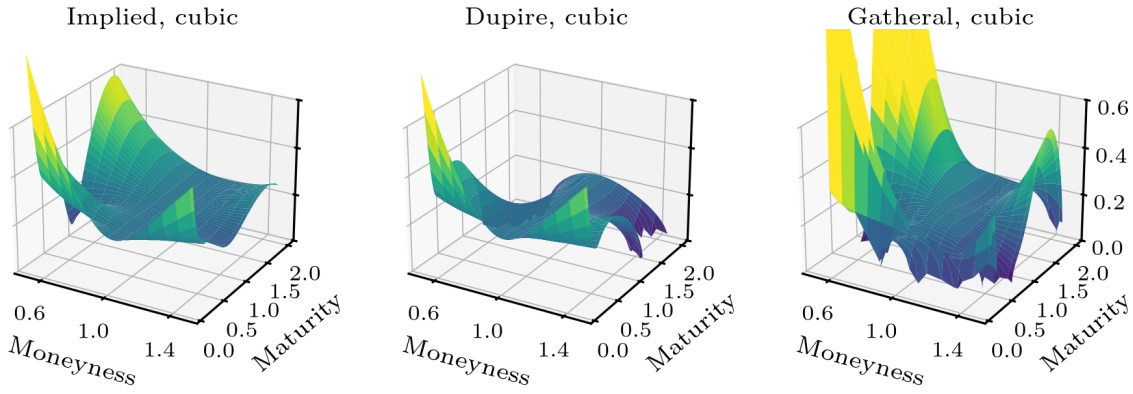
Nikkei 225 fitted surfaces. Blank regions carry no real volatility.

Appendix F.3. Nikkei 225 fitted surfaces, linear interpolation



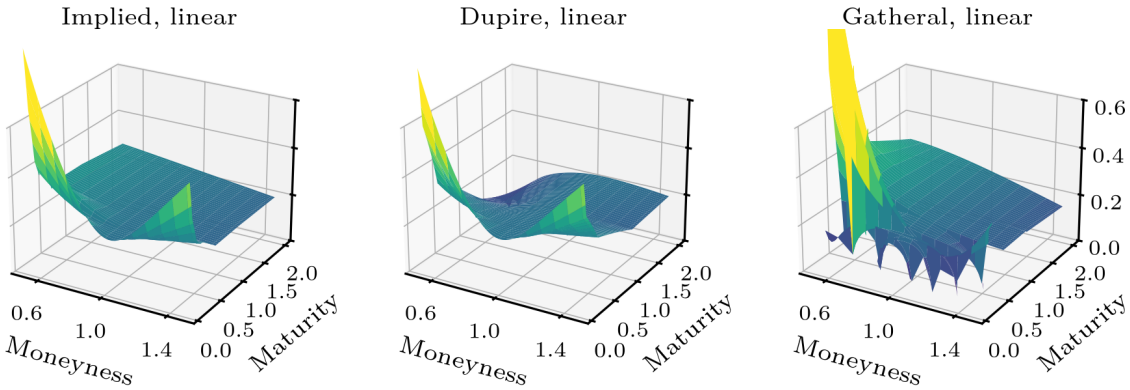
Nikkei 225 fitted surfaces. Blank regions carry no real volatility.

Appendix F.4. Hang Seng Index fitted surfaces, cubic interpolation



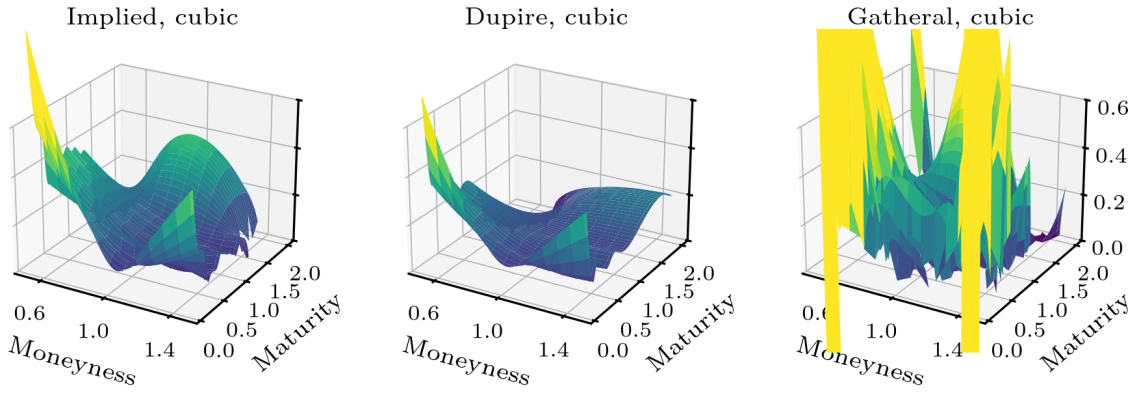
Hang Seng Index fitted surfaces. Blank regions carry no real volatility.

Appendix F.5. Hang Seng Index fitted surfaces, linear interpolation



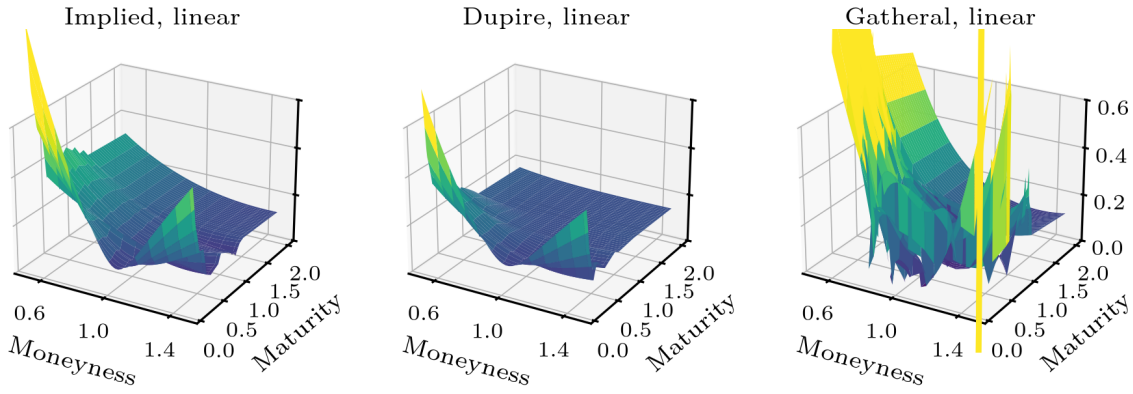
Hang Seng Index fitted surfaces. Blank regions carry no real volatility.

Appendix F.6. S&P 500 fitted surfaces, cubic interpolation



S&P 500 fitted surfaces. Blank regions carry no real volatility.

Appendix F.7. S&P 500 fitted surfaces, linear interpolation



S&P 500 fitted surfaces. Blank regions carry no real volatility.